

## Validation Lab

Separates the error we caused from the error we inherited - the single most useful diagnostic this project produced.

| PORT | PAGE FILE                     | MODULE                   | DESIGN                            |
|------|-------------------------------|--------------------------|-----------------------------------|
| 8503 | app/phase2/validation_page.py | phase2/validation/lab.py | shows the worst number on purpose |

Any team can report an error figure. The harder and far more valuable question is: whose error is it? Our model was trained to reproduce a reanalysis, so it can never be more accurate than that reanalysis. This page measures the reanalysis itself, and then subtracts.

### 1. What this feature is, in plain words

Our model learns from GLORYS, a physics-based ocean reanalysis. GLORYS is very good, but it is not perfect - it has its own error against real floats. If GLORYS is 1.1 degC wrong at a given depth, then a model that copies GLORYS perfectly is also 1.1 degC wrong there, and no amount of extra training will fix it.

So the Validation Lab measures three separate things and refuses to let their names blur together: **our model on real satellite input** (the actual deliverable), **our model fed reanalysis input** (its own ceiling), and **the reanalysis itself against floats** (the ceiling nobody else in this project had ever measured).

The result is a map of where effort would actually pay. It tells us which parts of the error are ours to fix and which are baked into the training target - and we publish both, including the parts that make us look worse.

#### THE EVERYDAY PICTURE

A student scoring 82% looks average until you learn the textbook they studied from has errors on 15% of its pages. This page grades the textbook first, then grades the student against what was actually learnable.

### 2. What you actually see on the screen

Three comparisons, deliberately never merged:

- Per-depth error for the model on satellite inputs - the deliverable, and the number that must be quoted.
- Per-depth error for the same model fed reanalysis inputs - a comparator only, never the headline.
- Per-depth error of the GLORYS reanalysis itself against independent Argo - the ceiling.
- A skill-versus-climatology chart that labels the crossover point outright, including the depth where climatology wins.
- The inherited-versus-ours split, using a stated tolerance that a reviewer can move.

### 3. How to run it, step by step

#### RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/validation_page.py --server.port 8503
```

- 1 Run the command and open **http://localhost:8503**.

- 2 Start on the per-depth error chart. Point out the bulge at 100 m - that is the thermocline, the hardest depth in the whole problem.
- 3 Switch on the reanalysis line. The audience sees our error and the training target's error almost touching through the thermocline.
- 4 Move to the inherited-versus-ours panel, which states which depths we could still improve.
- 5 Finish on the skill chart, and point at 1000 m where climatology beats us. Saying that out loud before a judge finds it is the most persuasive thing on this page.

#### REPRODUCE THE REANALYSIS CEILING

```
PYTHONPATH=src python scripts/phase2/glorys_vs_argo.py
```

## 4. Why the problem statement needs this

The problem statement asks for evaluation by RMSE, correlation and bias against independent observations. This page does that, and then does the thing the requirement implies but does not spell out: it establishes what a perfect answer would even look like given the training target available.

For a jury, it is also the credibility feature. A team that has measured its own ceiling and publishes the depth where a trivial baseline beats it is a team whose other numbers can be believed.

| PS requirement                     | How this feature answers it                                                                                    |
|------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Evaluate RMSE, correlation, bias   | All three, per depth, with the independent-Argo sample size shown beside every value.                          |
| Independent validation             | Scored against Argo profiles never used in training, matched through the Feature 2 engine.                     |
| Training target: GLORYS reanalysis | Measures that target's own accuracy, so the deliverable's error can be attributed rather than merely reported. |

## 5. The measured numbers behind it

The reanalysis was measured against independent floats with no model involved at all - 2,455 profiles reduced to 888 usable matches, 11,761 depth comparisons, within 5 days, distances max 18.7 km / mean 10.5 km / 95th percentile 16.5 km [VERIFIED]:

| Depth band              | Finding                                                                                                    | Whose error is it?                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| 100-150 m (thermocline) | The <b>Phase-1</b> model sat within <b>0.023 degC</b> of the reanalysis's own error                        | <b>Mostly inherited.</b> On the SHIPPED v2 model the gap at 100 m is <b>0.178 degC</b> - still mostly inherited, but not 0.023. |
| 20-50 m (mixed layer)   | <b>0.31 to 0.38 degC worse</b> than the reanalysis in Phase 1; <b>0.23 to 0.38</b> on the shipped v2 model | <b>Ours.</b> This is the one place effort would clearly pay.                                                                    |
| 1000 m                  | Climatology beats the shipped model by <b>0.055 degC</b>                                                   | Ours, small, and labelled on the chart. We win at 14 of 15 depths, not 15.                                                      |

- **These are the PHASE-1 figures, which is what this page displays.** The Lab reads the 2022 monthly record - 879 profiles, RMSE 0.9638 degC - not the v2 deliverable. Both columns are given above so the two are never confused: the inherited/ours split is qualitatively the same on v2, but the thermocline number is 0.178 degC, not 0.023.
- Tightening the match to 3 days and 25 km barely moved the reanalysis figures, which is how we know these are real reanalysis errors and not collocation mismatch.
- This single diagnostic redirected the project's remaining effort: a better mixed layer is available to us, a better thermocline is not.

## 6. The impact - and who it is for

| Who            | What changes for them                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------------------|
| The jury       | The clearest possible evidence of scientific honesty - a page built specifically to display the project's worst numbers. |
| Research teams | A method they can copy: measure your training target's own error before claiming your model's error is yours.            |
| INCOIS / MoES  | Realistic expectations. Knowing an error is inherited stops an agency funding work that cannot possibly succeed.         |
| The team       | A priority list backed by measurement rather than intuition.                                                             |

### WHY SOCIETY GAINS FROM THIS

Public money follows published accuracy claims. A system that distinguishes 'we could improve this' from 'nobody could improve this without better source data' lets funding go where it can actually change an outcome.

## 7. Say this to the jury

*This page exists to show you our worst numbers. Our model was trained on the GLORYS reanalysis, so we measured GLORYS itself against independent floats first. Through the thermocline most of our error is inherited - at 100 m the reanalysis scores 1.042 degC against the same floats while the shipped model scores 1.218, so 0.178 degC of that gap is ours. In the mixed layer we are 0.23 to 0.38 degC worse than the reanalysis, and that one is squarely ours. And at 1000 m plain climatology beats us by 0.055 degC, which we label on the chart. We beat climatology at 14 of 15 depths, not 15. The figures on the page itself are the Phase-1 measurement; the v2 equivalents are in the table.*

**Deliver this before the jury asks. Volunteering the weak number is the point.**

## 8. Questions the jury will ask

### Q. Is your model just copying GLORYS?

**A.** It is trained to reproduce GLORYS, so through the thermocline it is close to that ceiling - 0.178 degC above it at 100 m, against a reanalysis error of 1.042 degC at the same depth. But it does so from satellite surface inputs alone, which GLORYS does not use, and it runs daily on a 0.25 degree grid. Reaching the ceiling from cheaper inputs is the contribution.

### Q. Why is the thermocline the worst depth?

**A.** Because that is where a surface measurement constrains the water least. Our RMSE peaks at 1.22 degC at 100 m and correlation falls to 0.776 there. The reanalysis peaks at almost exactly the same depth, which is why we call that error inherited.

### Q. A crossover where climatology wins - is that not a failure?

**A.** It is a limit, at one depth of fifteen, worth 0.055 degC, and a test asserts the chart label appears on real data and does not appear when a model genuinely wins everywhere. We would rather be the team that labels it than the team that is caught by it.

## 9. The honest limit - say it before they find it

### A DISAGREEMENT THAT WAS OPEN, NOW TRACED

The Validation Lab headline (0.9638 degC over 879 profiles) and the freeze manifest (0.9006 degC over 962 profiles) are two different models on two different records: **lab.py reads artifacts/argo\_error\_by\_depth.json**, the Phase-1 satellite-driven model scored against 2022 floats on the monthly grid, while the manifest records the v2 daily model against 2025-26 floats under the seafloor\_masked\_v1 protocol. Confirmed by reading the code on 2026-09-07, not inferred.

#### IN ONE LINE

*The page that measures our training target's own error first, so the deliverable's error can be attributed instead of merely reported.*

**0.178 degC**

OF THE THERMOCLINE ERROR THAT IS OURS,  
NOT THE TARGET'S

**+0.23 to +0.38**

DEGC THAT ARE GENUINELY OURS, IN THE  
MIXED LAYER

**14 of 15**

DEPTHS WHERE WE BEAT CLIMATOLOGY

Feature 3 of 17. Next: Feature 4.